

ADITYA SURESH KAMATH

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SUMMARY

Machine Learning Engineer with 3 years of production experience designing production grade Generative AI and agentic AI systems, including a conversational AI platform serving 10K+ daily users at 99.9% uptime. Experienced in Retrieval-Augmented Generation architectures, multi-agent orchestration, model context protocol (MCP) and scalable production deployment.

EDUCATION

University of Illinois Urbana-Champaign <i>Master of Science in Information Management (GPA: 4.0/4.0)</i>	Aug 2025 – May 2027 (Expected) Champaign, IL
SVKM's NMIMS University <i>B.Tech. in Electronics and Telecommunication (GPA: 3.94/4.0)</i>	Jul 2018 – Aug 2022 Mumbai, India

WORK EXPERIENCE

Rivian <i>Data Science - Advanced Diagnostic Integration Intern</i>	May 2026 – Present Illinois, USA
Reduced vehicle fault investigation time by 70% by building fault-finder, a Python-based Model Context Protocol (MCP) server using FastMCP that automated diagnostics across Jira, GitLab, Databricks, and Google Cloud Firestore. - Improved diagnostic accuracy of AI-assisted fault investigation by designing a persistent memory architecture with 3 collections (cases, corrections, tips), supporting JSON and Databricks Delta Lake storage backends.	
Symmetric IT Services <i>Senior Machine Learning Engineer</i>	Dec 2024 – Jun 2025 Mumbai, India
- Architected and productionized a client-facing insurance claims copilot using LangGraph, ChromaDB, and FastAPI; integrating RAG retrieval, episodic memory, and tool-based actions with a human-in-the-loop approval layer; containerized with Docker and deployed via CI/CD on GitHub Actions and AWS EC2 - Built 6 n8n automation workflows including a document ingestion and summarization pipeline and a RAG-driven project retrieval system, cutting internal processing time from 3-4 hours to under 10 minutes	
Quantiphi <i>Machine Learning Engineer</i>	Aug 2022 – Nov 2024 Mumbai, India
- Architected a GKE-based, distributed conversational AI platform with 85 routing endpoints, sustaining 10K+ daily users at 99.9% uptime across cloud-native infrastructure Trained and Fine-tuned a BERT-based sentence transformer (NLP) on 500K domain-specific records and replaced Redis calls with full in-memory caching, cutting inference latency by 62%. - Built a custom topic modeling pipeline using LDA and Top2Vec on 10M+ records, producing 160 topic categories at 88% accuracy, outperforming Google's native Contact Center AI model in both coverage and precision - Engineered BigQuery data pipelines (ETL) processing 1M financial records via Cloud Run to power Looker Studio dashboards with automated alerting; reduced issue detection time from 4 days to 5 hours, surfacing a billing error in user segment	

PROJECTS

CropSense - AI-powered crop disease diagnosis
- Developed a multimodal crop disease diagnosis computer vision system using VGG16 CNN (89% accuracy), RAG based retrieval, and a multi-LLM consensus pipeline to analyze crop images, farmer voice input, and geolocation weather data - Delivered confidence-scored recommendations through a FastAPI backend and Flutter app with TTS and PDF report generation
CareCore – Agentic automation for clinical operations
- Implemented a 3-agent clinical care plan copilot that pulls patient data from Dataverse, summarizes medical history and prescriptions, and auto-fills official care plan templates for human review - Cut drafting time from 4 hours to under 10 minutes by eliminating manual data collection with enforced guardrails and human-in-the-loop approval to ensure clinically grounded, auditable outputs

TECHNICAL SKILLS

- Languages:** Python, SQL, R
- GenAI & LLM:** LangChain, LlamaIndex, RAG Pipelines, Prompt Engineering, Fine-tuning, LLM Evaluation, MCP, A2A
- Data & Databases:** Apache Spark, BigQuery, Airflow, dbt, PostgreSQL, ChromaDB, Pinecone, Supabase
- Developer Tools:** Docker, Kubernetes, Vertex AI, SageMaker, CI/CD, MLOps, Pytest, Tableau, Power BI, Looker Studio
- Cloud:** Google Cloud Platform, Amazon Web Services, Microsoft Azure, Databricks
- GCP Certifications:** Google Cloud Professional Machine Learning Engineer, Google Cloud Professional Cloud Architect

LEADERSHIP SKILLS

Top 3 Finalist, UIUC Research Park Intern Awards, 2026 | Tech Lead, Applied Business Research